

Market Analysis of Airbnb in D.C. Area

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Summary of Findings

An analysis of two datasets produced a dataframe consisting of 1,931 rows and 15 columns were examined. Price outliers were removed resulting in a 1,842 row by 17 column datasets. A number of analyses were performed on the Airbnb housing data in the Washington, D.C. area. Correlation of the numeric variables were analyzed and found accommodates, bedrooms, and bathrooms had the highest correlation with price. Additionally, the simple linear regression model found that bathrooms had the smallest residual standard error (SER). To explore the data further, the multiple regression models were performed using room_type, neighborhood, bathrooms, and bedrooms as predictors for the response variable price. From this model, 41.6% of the variation in price was explained by these predictors. Our analysis found that entire homes/apartments in the Downtown, Foggy Bottom, and Georgetown neighborhoods are areas with the highest growth potential.

Methods

Question 1

Exploratory data analysis was performed on two datasets: Listings.csv and Reviews.csv. A Left Join on id and listing_id was performed, resulting in a dataframe with 1,931 rows and 15 columns. NAs were found in the beds and avg_rating columns and were replaced with average of beds (rounded up) and average avg_rating, respectively. Blank values were observed in the room_type column and were replaced with "Unknown." Once the dataframe was clear of NAs and blank values, the data types of each column were examined. It was determined that neighborhood, room_type, bathrooms should be converted to factors and host_since be converted to a date data type.

Avg_rating and host_acceptance_rate were left-skewed while total_reviews and host_total_listings were right skewed. For the left skewed data, the complement of the value was taken followed by a square root transform or a log transform was performed; however, these were not successful in normalizing the data. Additionally for the right skewed data, a log and square root transform was performed, which was also unsuccessful in normalizing the dataset.

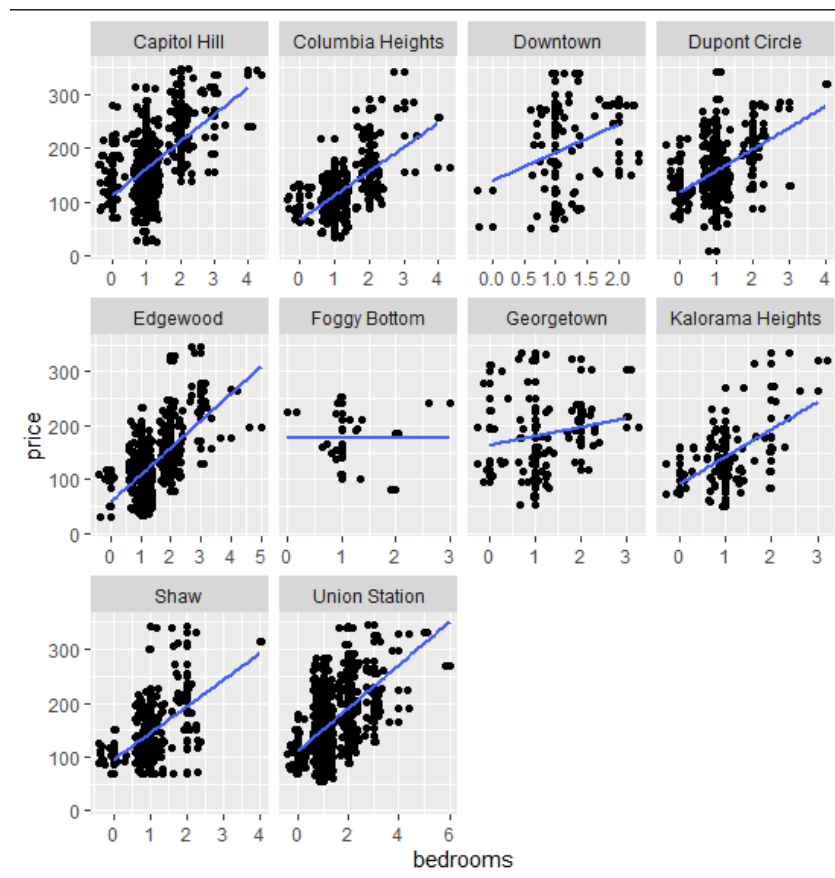
Given that the dataset was skewed, the interquartile range (IQR) method was used in favor of the z-score method. This method was used to determine the outliers for numerical variables. A correlation table was performed on the numeric variables observing the response variable price. It was found prior to filtering outliers for price, host_total_listings are the most correlated with price. Outliers were removed from price using the IQR method and then a correlation table was performed again. It was found that accommodates, beds, and bedrooms had the highest correlation. To optimize the dataset further, outlier analysis was performed using accommodates, beds, and bedrooms. A correlation table was performed for each combination and it was determined that removing the price outliers was the most effective. The resulting dataset results in a 1,842 row by 17 column dataframe.

	price	price_acc\$price	price_br\$price	price_bed\$price
accommodates	0.465	0.449	0.454	0.440
bedrooms	0.449	0.404	0.428	0.392
beds	0.316	0.271	0.291	0.344
host_total_listings	0.226	0.241	0.233	0.244
host_acceptance_rate	0.163	0.158	0.161	0.164
avg_rating	-0.0195	-0.0275	-0.0161	-0.0421

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superhost2	-0.0369	-0.0500	-0.0345	-0.0629
total_reviews	-0.126	-0.121	-0.122	-0.133
min_nights	-0.113	-0.138	-0.133	-0.142

With the accommodates, bedrooms, and beds predictors having the highest correlation to the response variable price, a further drill down analysis was performed to identify the relationship between these variables segmented by neighborhood type. The scatterplots for each of the three variables were performed, and it was determined that the bedrooms had the most interesting relationship to price as shown in the figure below. It was found that as the number of bedrooms increased a positive linear relationship was observed in all the neighborhoods, except for the Foggy Bottom neighborhood which had a near zero correlation.



Scatterplot of Price Versus Bedrooms Segmented by Neighborhood

Question 2

In the original dataset, the neighborhood and room type combination of Kalorama Heights and Hotel Room had the highest average price with a value of \$50,000. Given that this value is two orders of magnitude greater than the other data points, this calculation and the following analysis was redone using the dataset with the price outliers trimmed. It was determined that in this dataset the combination of Foggy Bottom with unknown room type had the highest average price of \$241; however, this combination had a sample size of one. Downtown entire home/apt was the third most expensive, based on average price with a value of \$226 but had a larger sample size of 45. The combination with the lowest average price was found to be shared rooms in the Capitol Hill neighborhood with an average price of \$54.3.

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The variance of the price was studied for the neighborhood and room type combinations and it was found that the Downtown private rooms had the highest standard deviation with a value of 95.1 and the Downtown shared rooms had the lowest standard deviation with a value of 8.33. Given that this data set was slightly skewed, the analysis of variance was also studied using the IQR, which found that the combination of Downtown private rooms still had the highest variance with an IQR of 192. Due to the limited sample size of Capitol Hill hotel room, Downtown hotel room, Foggy Bottom Unknown, and Kalorama Heights shared rooms all had a sample size of 1 which resulted in an IQR of 0. The lowest nonzero IQR was found to be the Downtown shared room with an IQR of 8. Based on the variance analysis using both the standard deviation and IQR methods, we concluded that the Downtown private room had the highest variance, while the Downtown shared room had the lowest variance.

Question 3 / Question 4

While constructing a function to consider a confidence level and the price of Airbnb's in the DC area that were included in our dataframe, our analysis found, with a 95% confidence interval for the average listing prices of the dataset, the following:

Lower	Mean	Upper	n
152.36	155.29	158.23	1,842.00

We are 95% confident that the true population mean lies between \$152.36 and \$158.23 per night. To test this, analysis on whether the average of all listings in the population is more than \$200, at a 95% confidence level, produced the following:

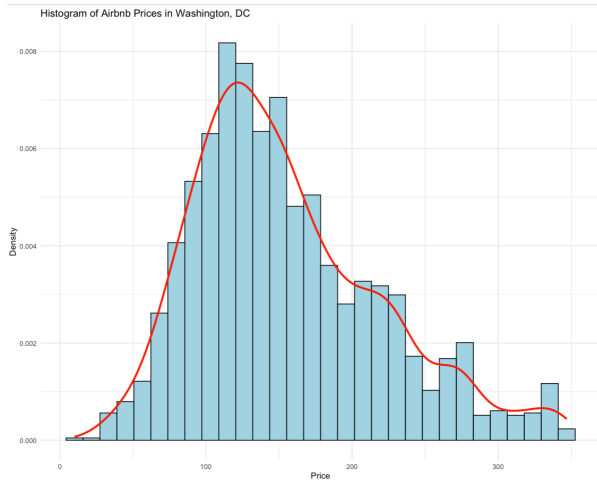
t	df	p-value
-29.848	1,841	1
Alternative Hypothesis: true mean is greater than 200		
95 percent confidence interval:	152.8294	Inf
Sample estimates: mean of x: 155.2942		

Given our analysis, we are unable to reject the Null Hypothesis given that the p-value is not less than 0.05.

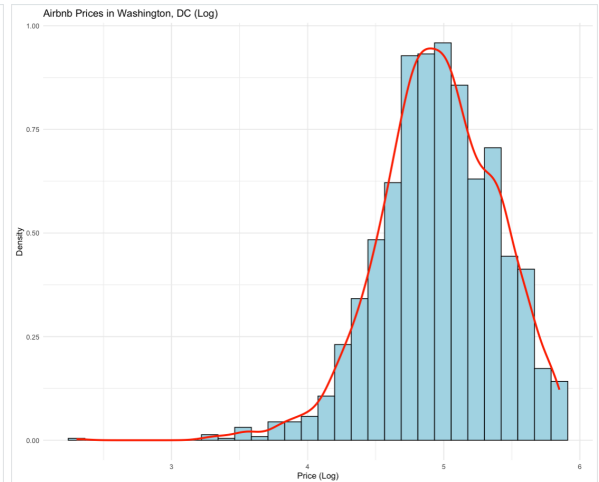
Question 5

In testing for normality of the Airbnb listing prices, a histogram was generated as the visual. Since the outliers were removed from the dataset, we did not have to account for prices that were extremely high or low. The histogram shows that the distribution of the raw price is right-skewed, or positively skewed. Most of the listings are priced between \$75 and \$150 with a few listings having price increases beyond \$200. This indicates that while most Airbnb listings in the DC neighborhood are moderately priced, a few high-priced listings pull the mean forward, thus suggesting that the data are not normally distributed.

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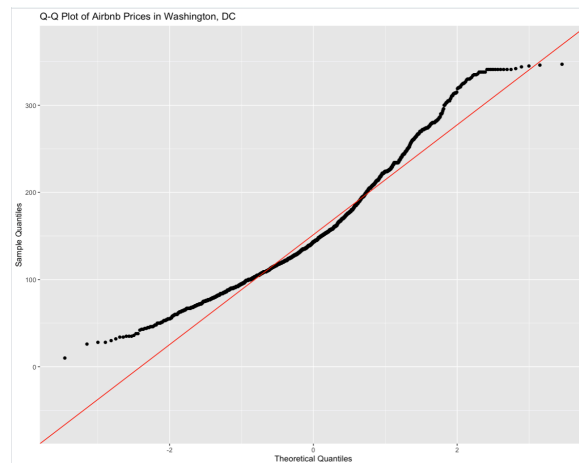


Histogram with Raw Price



Histogram with Log Transformation

When applying the log transformation to the price, the distribution becomes more symmetric. We used $\log(\text{price})$ in an attempt to reduce the skewness. Applying the log flipped the price distribution, showing a left-skewed model. We concluded that by removing the outliers in price at the beginning, the distribution is more normalized.



Q-Q Plot with Raw Price

To support the normality test with the histogram, we generated a Q-Q plot with the raw price. The red line signifies the normal distribution. The Q-Q plot confirms that the prices deviate from normality, particularly at higher values. The points at the right tail at the upper end deviate from the red line, showing the heavier tails while the middle points follow the red line fairly well. The Q-Q plot and histogram both show that high-priced Airbnb listings are much higher than what a normal distribution would predict.

Question 6

Using the simple linear regression model, we modeled various predictors for the listing prices based on R-squared and residual standard error (SER).

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	Model	SER	R2
1	Reg with Neighborhood	61.25694	0.0963342077
2	Reg with Bedrooms	57.45585	0.2015308358
3	Reg with Room Type	60.89715	0.1044808891
4	Reg with Accommodates	56.91354	0.2165326508
5	Reg with Beds	61.01476	0.0995503403
6	Reg with Bathrooms	55.16945	0.2686162663
7	Reg with Min Nights	63.72875	0.0176632197
8	Reg with Total Reviews	63.78832	0.0158257530
9	Reg with Host Acceptance Rate	63.44086	0.0265182301
10	Reg with Superhost	64.25538	0.0013607687
11	Reg with Host Total Listings	62.63565	0.0510729702
12	Reg with Average Rating	64.28685	0.0003821928

```
>
> #Model with lowest residual standard error
> modelComp$Model[which.min(modelComp$SER)]
[1] "Reg with Bathrooms"
>
> #Model with highest residual standard error
> modelComp$Model[which.max(modelComp$SER)]
[1] "Reg with Average Rating"
```

Comparison Table for Predictors of Listing Prices

In comparing the results, we found that the best simple linear regression model for predicting price is bathrooms. It has the lowest residual standard error, SER = 55.16945, and the highest R^2 at 0.2686162663. At 55.17%, bathrooms have the best prediction accuracy and explains the most variation in price at 26.8%. Alternatively, the worst simple linear regression model for prediction price is the average rating. It has the highest residual standard error at 64.28685.

In conclusion, the comparisons of the simple linear regression models show that bathrooms and bedrooms have the strongest statistical relationships when it comes to Airbnb listing prices. The average ratings do not have a significant effect on the price.

Question 7

Using a multiple linear regression, we modeled price against neighborhood, room type, bedrooms, and bathrooms that covers a sample size of 1,842 listings. From the coefficients, the baselines are Capitol Hill (neighborhood) and Entire home / apt (room_type). The model explains roughly 42% of price variation (Mult. R^2 : .4252, Adj. R^2 : .4166), with a residual standard error of \$49. Diagnostics show the relationship is linear, with variances increasing slightly at higher prices.

Most of our independent variables are statistically significant, which is aligned with pricing. Larger listings with full bathrooms located in Downtown achieve the highest prices. In contrast, listings in Columbia Heights, Edgewood, or listings with shared rooms / baths are at the lower pricing spectrum.

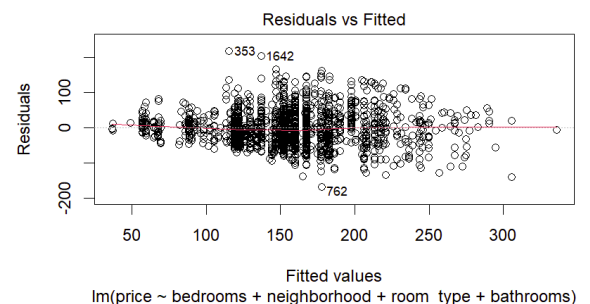
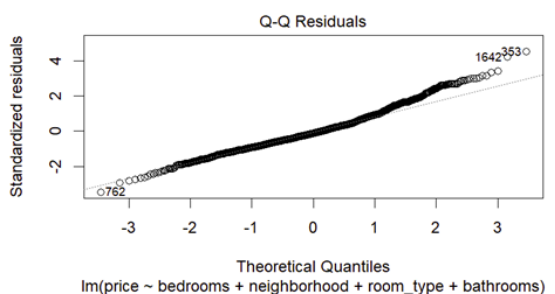
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	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	136.913466	3.643874	37.5736003	5.064628e-229
bedrooms	30.277802	2.206155	13.7242379	7.411175e-41
neighborhoodColumbia Heights	-47.954089	4.424933	-10.8372460	1.463895e-26
neighborhoodDowntown	38.983961	6.639195	5.8717907	5.115536e-09
neighborhoodDupont Circle	-7.558652	4.498532	-1.6802484	9.308117e-02
neighborhoodEdgewood	-45.563815	4.146886	-10.9874776	3.103711e-27
neighborhoodFoggy Bottom	10.323697	11.358598	0.9088884	3.635298e-01
neighborhoodGeorgetown	10.237417	6.373672	1.6062040	1.084031e-01
neighborhoodKalamazoo Heights	-20.573244	6.233747	-3.3003015	9.845337e-04
neighborhoodShaw	-11.008809	4.965202	-2.2171927	2.673348e-02
neighborhoodUnion Station	-15.004493	3.678726	-4.0787196	4.724493e-05
room_typeHotel room	-10.960337	14.342893	-0.7641650	4.448683e-01
room_typePrivate room	-19.257899	7.205703	-2.6725914	7.594204e-03
room_typeShared room	-55.092738	16.360196	-3.3674864	7.744231e-04
room_typeUnknown	-2.340214	8.527619	-0.2744276	7.837873e-01
bathrooms1 private bath	-13.912133	8.394341	-1.6573227	9.762717e-02
bathrooms1 shared bath	-42.765915	9.498027	-4.5026104	7.139478e-06
bathrooms1.5 baths	18.246978	6.075214	3.0035122	2.705343e-03
bathrooms1.5 shared baths	-34.622358	11.614172	-2.9810441	2.910910e-03
bathrooms2 baths	31.057706	5.012446	6.1961173	7.144449e-10
bathrooms2 shared baths	-35.177655	11.926248	-2.9495995	3.222599e-03
bathrooms2.5 baths	36.829294	7.519079	4.8981124	1.053567e-06
bathrooms2.5 shared baths	11.764692	17.893908	0.6574691	5.109627e-01
bathrooms3 baths	62.528593	13.705030	4.5624556	5.394724e-06
bathrooms3 shared baths	-65.796856	21.412718	-3.0727932	2.152204e-03
bathrooms3.5 baths	-6.552119	25.485380	-0.2570932	7.971360e-01
bathrooms4 baths	-26.297980	35.844889	-0.7336605	4.632505e-01
bathroomsShared half-bath	-66.928875	49.678416	-1.3472425	1.780704e-01

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The residual vs. fitted plot confirms our model as a linear relationship. That said, there is a slight variance in residuals at the higher end of pricing, suggesting variability increases with price. The Q-Q plot sticks closely to the line confirming normality, with similar variability identified in the residual vs. fitted plot. Overall, the linear regression holds up well with no single variable we can identify affecting the model significantly.



Our findings suggest that larger properties, more bedrooms and additional full bathrooms command significantly higher prices, while listings with shared bathrooms or small layouts rent for less. Premium neighborhoods, particularly Downtown continue to outperform the baseline (Capitol Hill), while Columbia Heights and Edgewood remain lower valued. Entire home / apartment listings still achieve the highest returns, outperforming private and shared rooms by significant margins. The Top-5 Avg. predicted price results confirm these patterns, emphasizing Downtown Entire / Homes as the most profitable segment.

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Question 8

Multicollinearity is not a major concern in this model. Since there is only one numeric predictor (bedrooms), collinearity between continuous variables does not apply. The remaining variables are categorical (room_type, bathroom, and neighborhood), which are represented by dummy variables in the regression model. That said, we cannot entirely rule out that categorical variables may introduce some collinearity, however, given the fit of the model, possible effects are not believed (in this case) to materially impact overall results.

Recommendations

The Washington, D.C. area is an urban center often attracting tourists visiting the various attractions in the nation's capital or visiting on a business trip. Airbnb aims to have an ideal room for any situation a visitor may have. It is our recommendation that Airbnb prioritize pricing strategies in neighborhoods where price predictions are strongest. Our analysis identified Downtown, Foggy Bottom, and Georgetown as high performing neighborhoods, where listings that featured more bedrooms and full bathrooms correlate with higher nightly rates. These neighborhoods represent a great opportunity for targeted growth. Entire home/apartment listings remain the most profitable segment as they dominate the top 5 price rankings. Pricing in these categories ranged from \$178 to \$224 per night on average. Four of the top five categories were entire home/apartment properties, underscoring the market preference for larger units. For underrepresented neighborhoods such as Foggy Bottom (20 listings in total), where we know average pricing is in the top 5, we recommend Airbnb consider focusing marketing efforts to appeal to homeowners within that particular segment of the market.

The current model explains approximately 41.6% of price variation in the D.C. area, which can be further improved by creating a more comprehensive dataset. It is advisable that further analysis on the square footage of the properties, property ownership type (owner-occupied vs. investor-owned), and the distance from tourism attractions can be studied to further improve the model. Incorporating these features would improve the accuracy of predictive pricing models and provide Airbnb with a more precise framework for dynamic pricing and localized marketing strategies.